Project Title

UK Housing Affordability and Regional Rental Analysis (2015–2025)

Project Summary

This Power BI project investigates UK housing affordability and rental market trends between 2015 and 2025. The analysis combines rental and income datasets to evaluate how rental costs have changed over time, whether income growth has kept pace with housing costs, and how affordability varies across regions and property types.

The report is divided into two dashboards:

1. Housing Affordability Analysis
2. Regional Rental Analysis

The dashboards provide an interactive view of rental trends, income growth, affordability pressures, regional rental disparities, and property-type performance.

Data Source

The project uses UK housing and income datasets covering the period 2015–2025.

The datasets contain information related to:

- Average rental prices
- Rental prices by property type
- Public sector earnings
- Private sector earnings
- Total earnings
- Geographic areas
- Yearly housing indicators

Data Preparation

Before importing the datasets into Power BI:

- Unnecessary columns were removed in Excel.
- Only business-relevant fields were retained.
- Dataset structure was simplified to improve readability and performance.

Within Power BI:

- Date Table was created.
- Area Table was created.
- Relationships were established between tables.
- Property type columns were unpivoted in Power Query.
- DAX measures were created for affordability, income, rankings, and KPIs.

Data Model

Fact Tables

Rent

Contains:

- Year
- Area Name
- Property Type
- Property Rent
- Average Rent

Income

Contains:

- Year
- Public Pay
- Private Pay
- Total Pay

Dimension Tables

DateTable

Created to:

- Enable time-based analysis
- Support filtering across visuals
- Ensure consistent year relationships

AreaTable

Created to:

- Support area slicers
- Improve model structure
- Enable area-based filtering

DAX Measures Used

Average Rent
Avg_Rent =
AVERAGE(Rent[Avg Rent])

Average Income

Avg_Income =
AVERAGE(Income[Real Total Pay Whole])

Monthly Income

Income data was recorded weekly, so it was converted to monthly values.

Monthly Income =

[Avg_Income] * 4.33

Public Sector Pay

Avg Public Pay =

AVERAGE(Income[Total Pay Public]) * 4.33

Private Sector Pay

Avg Private Pay =

AVERAGE(Income[Total Pay Private]) * 4.33

Latest Average Rent

```
Latest Avg Rent =

VAR LatestYear =

MAX(Rent[Year])


RETURN

CALCULATE(

  [Avg_Rent],

  Rent[Year] = LatestYear

)
```

Latest Monthly Income

```
Latest Monthly Income =

VAR LatestYear =

MAX(Income[Year])


RETURN

CALCULATE(

   [Monthly Income],

   Income[Year] = LatestYear

)
```

Affordability Percentage

```
Affordability % =

DIVIDE(

   [Latest Avg Rent],

   [Monthly Income]

)
```

Top 10 Area

```
Top 10 Area =

IF([Area Rank] <= 10,1,0)
```

Average Property Rent

```
Avg Property Rent =

AVERAGE(Rent[Property Rent])
```

Dashboard 1: Housing Affordability Analysis

Purpose

To evaluate the relationship between rental costs and income growth and assess affordability over time.

KPIs

Latest Average Rent

Current average monthly rent.

Latest Monthly Income

Current average monthly income.

Affordability %

Percentage of income required to pay rent.

Income vs Rental Trend

Shows how rent and income evolved over time.

Insight

Rental prices and income increased throughout the period, although rental costs accelerated more rapidly after 2022.

Housing Affordability Trend

Tracks the percentage of income spent on rent.

Insight

Affordability remained relatively stable before 2021 but deteriorated significantly during later years.

Year-over-Year Growth %

Compares annual rent growth against income growth.

Insight

Rental cost growth exceeded income growth in several recent years.

Dashboard 2: Regional Rental Analysis

Purpose

To investigate geographical rental differences and property-type trends.

Top 10 Most Expensive Rental Areas

Ranks areas based on average rental prices.

Insight

London boroughs dominate the highest-rent locations.

Public vs Private Sector Pay Trend

Compares public and private earnings.

Insight

Both sectors experienced long-term growth, with private earnings increasing slightly faster in later years.

Property Type Rent Trends

Compares rental trends across:

- Detached
- Semi-Detached
- Flats
- Terraced

Insight

Detached properties consistently recorded the highest rents, while flats remained the least expensive.

Rental Trend for Selected Area

Provides area-specific rental analysis through slicers.

Insight

Rental growth patterns vary significantly across regions.

Challenges Faced

Challenge 1

Blank KPI values.

Solution

Created latest-year measures using CALCULATE and filtering logic.

Challenge 2

Blank year issues in visuals.

Solution

Verified DateTable relationships and ensured visuals used DateTable fields.

Challenge 3

Property Type Slicer Not Possible

Cause

Property types existed as separate columns.

Solution

Unpivoted property-type columns into:

- Property Type
- Property Rent

Challenge 4

Top 10 Area Filtering

Solution

Created Area Rank and Top 10 Area measures.

Challenge 5

Weekly Income Values

Solution

Converted weekly earnings into monthly values using a 4.33-week multiplier.

Key Business Insights (Dashboard 1)

- UK rental prices showed a long-term upward trend between 2015 and 2025 despite some short-term fluctuations.
- Income growth did not fully keep pace with rising housing costs.
- Housing affordability deteriorated significantly after 2021.
- Rental cost growth accelerated faster than income growth in recent years.

Key Business Insights (Dashboard 2)

- London boroughs consistently recorded the highest rental prices.
- Detached properties remained the most expensive property type throughout the analysis period.
- Rental price differences between regions became more noticeable over time.
- Public and private sector wages increased steadily across the analysis period.